

The Embodiment Trap: When AI Has a Body in Education

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Abstract

Artificial Intelligence (AI) is increasingly imagined, designed, and prototyped in embodied forms (e.g., avatars or virtual humans). Within HCI and educational technology, embodiment is commonly framed as a means to improve engagement, presence, or motivation. We argue that such framings are incomplete for educational settings. When AI is given a body in education, embodiment functions as a claim to legitimacy that redistributes pedagogical authority, often without explicit scrutiny. Drawing on prior work in embodied interaction, social presence, AI in education, and critical computing, we identify what we call *the embodiment trap*; a field-level tendency to treat bodily presence as a design benefit while sidelining its normative consequences. We articulate four tensions introduced by embodied AI in education and examine why these tensions persist. With this paper, we invite the HCI community to reconsider embodiment not as an interface enhancement, but as a normative intervention shaping educational roles, responsibilities, and futures.

Keywords

extended reality, education, AI, embodiment, embodied AI

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1 Introduction

Education has never been only about information transfer. It is a relational practice through which authority is enacted, trust is negotiated, and responsibility is assigned [31]. Educational technologies, from blackboards to projectors to learning management systems, have always participated in this process by reorganizing who speaks, who listens, and whose knowledge counts [15, 22, 30].

Artificial Intelligence (AI) is extending this lineage by introducing new forms of presence and participation in educational contexts. Advances in generative AI and extended reality (XR) allow AI systems to appear not only as tools or background services, but as embodied figures (e.g., avatars, virtual humans, and immersive companions co-present with learners) [23, 40]. These systems can gesture, speak, respond contingently, and occupy space alongside students and educators [18]. Prior work shows that such embodiment fundamentally alters how systems are interpreted, increasing social attribution, perceived credibility, and compliance, even when users are fully aware that the system is artificial [12, 18, 27, 39].

Commercial XR platforms are also beginning to market AI-powered characters or avatars for learning and training. For example, ENGAGE XR positions itself as an education-oriented social VR platform [10], and highlights generative-AI features that let users interact with AI-powered avatars; it also showcases a “School of AI” experience where learners can converse with historical characters in an immersive environment. Fectar is an AR/XR creation and viewing platform that documents how creators can add an AI avatar (virtual assistant) inside an XR “space” (e.g., an avatar that can “speak and explain on tap”), which is framed as useful for educational experiences, demos, and tours [11]. Similarly, systems such as XR4ED [18, 37] explore extended reality educational platforms that include virtual teacher interactions and AI-enhanced guidance features.

However, embodied AI in education is frequently framed in terms of technical enhancement and experiential gains. Research on embodied pedagogical agents, social robots, and XR-based tutors often evaluates such systems through improvements in engagement, presence, motivation, or perceived learning support [1, 18, 19, 23, 39, 40]. Embodiment is therefore commonly justified as a means of increasing immersion or instructional effectiveness, rather than as a normative intervention into pedagogical authority. With this paper, we challenge that framing. Embodiment is not a neutral design choice. A body claims attention, invites trust, and signals authority. In educational contexts, where the figure of an educator carries deep cultural and moral significance, these signals are especially consequential. We aim to advance a provocation directed at the HCI community, and open up a discussion that when we give AI a body in education, we are not merely designing better interfaces but rather we are redistributing pedagogical authority.



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2 Embodiment as Authority

HCI research has long emphasized that embodiment is relational rather than representational. Dourish’s theory of embodied interaction positions embodiment as a way in which people make meaning with and through technology [8]. Extending this relational view, Suchman shows that technologies acquire agency and authority not through intrinsic intelligence, but through situated interpretation within social and institutional contexts [32]. From this perspective, embodiment is not merely about giving technology a particular form or appearance; it becomes a mechanism through which legitimacy is interpreted and enacted.

Empirical work reinforces this interpretation. Media equation research demonstrates that people respond socially to even minimal cues of agency, treating systems as intentional and socially capable despite explicit knowledge that they are machines [29]. Studies of pedagogical agents and social robots show that embodied systems are often perceived as more knowledgeable, trustworthy, and authoritative than disembodied alternatives, even when underlying capabilities are identical [1]. Importantly, these effects persist across levels of realism. This suggests that authority does not depend on human-likeness alone, but on presence and participation.

In educational settings, embodiment intersects with deeply entrenched cultural expectations about teaching and expertise. According to Legitimate Peripheral Participation and Communities of Practice [38], authority and legitimacy are not merely individual attributes but are negotiated within communities of practice; embodied AI may therefore be read as occupying positions traditionally reserved for legitimate participants in teaching roles [17]. Classrooms are historically organized around visible authority (e.g., who stands, speaks, evaluates, and corrects). The physical presence of educators is therefore central to how authority is recognized and sustained. As a result, when AI systems appear embodied in these spaces (e.g., co-present with students, responsive in real time, and positioned as sources of guidance), they are read through the same interpretive frames as human instructors [8]. This creates a subtle but consequential shift. Embodied AI does not merely deliver information; it participates in the social choreography of teaching. It gestures, addresses learners directly, and occupies shared space. In doing so, it signals entitlement to guide, evaluate, and intervene. Crucially, this signaling operates independently of designers’ intentions. Even systems framed as “assistive” may be interpreted as authoritative simply by virtue of how and where they appear.

From this perspective, embodiment functions as a claim to legitimacy. Once a system is embodied, its outputs are no longer encountered as neutral suggestions or background support, but as situated acts performed by a visible agent. This claim cannot be easily revoked through interface cues, transparency statements, or disclaimers. Authority, once enacted, is difficult to contain [2]. In the education domain, these dynamics are amplified because of ethical implications. Educators are accountable for learners’ development, model epistemic humility, and negotiate uncertainty and care [16]. Embodied AI systems, however, cannot bear these responsibilities, even as they may appear to perform them. The tension, therefore, is not whether AI can act like a teacher, but whether it should be allowed to appear as one.

3 The Embodiment Trap

We argue that HCI and educational technology research are drifting into what we call the *embodiment trap*. The embodiment trap arises when the presence of a body is treated primarily as a benefit (e.g., something that improves engagement, immersion, or usability), while its normative consequences are underexamined.

Within HCI, embodiment has often been associated with progress such as richer interaction and more “natural” communication. In education, these assumptions are reinforced by institutional pressures to increase engagement, motivation, and scalability. As a result, embodied AI is commonly evaluated through experiential metrics (e.g., attention, satisfaction, perceived usefulness) [21, 26], while its effects on authority, responsibility, and power remain peripheral. Framed as an unambiguous improvement, embodiment becomes a default design trajectory rather than a contested choice.

The embodiment trap mirrors critiques raised in other domains of AI deployment [24], where optimization-centric narratives align with technical and institutional incentives, even when they undermine human agency and accountability [6]. In education, this may obscure how embodied AI reconfigures pedagogical authority. Importantly, the embodiment trap is not the result of malicious intent or careless design. It emerges from well-meaning efforts to make AI more accessible, more intuitive, and more effective, especially when combined with XR [20]. Yet, these efforts can inadvertently stabilize problematic assumptions; for example, that increased presence is always desirable, and that authority can be safely delegated so long as systems perform well.

HCI has a history of interrogating its own assumptions when technologies begin to (re)shape social roles and institutional power. Embodied AI in education demands a similar reflexivity. The danger is not that embodied AI will fail, but that it will succeed in performing authority without responsibility in an attempt to appear competent, attentive, and reliable while remaining fundamentally unaccountable. With this provocation, we aim to shift attention from isolated systems to field-level patterns. The question is not whether individual embodied AI applications are well designed, but whether HCI is collectively normalizing a redistribution of pedagogical authority without adequate debate.

4 Four Tensions We Cannot Design Away

We articulate four tensions that accompany embodied AI in educational contexts. These tensions are not empirical findings or design trade-offs to be resolved through better interfaces. Instead, they reflect structural contradictions that emerge when AI systems are granted bodily presence in environments historically organized around human authority, care, and responsibility. Together, these tensions shift the question from whether embodied AI works to whether it is legitimate in educational contexts. Table 1 summarizes the four tensions and clarifies what embodiment enables, what it risks normalizing, and what is at stake.

4.1 Presence vs. Authority

Embodied presence is often celebrated for its ability to foster co-presence, attention, and engagement [3]. In learning environments, presence is associated with immediacy, responsiveness, and pedagogical effectiveness [34]. However, presence is never merely

Table 1: Four tensions introduced by embodied AI in educational contexts. These tensions illustrate how bodily presence simultaneously enables interactional benefits while reshaping authority, responsibility, and educational roles.

Tension	Embodiment Benefits	Embodiment Risks	What is at stake
Presence vs. Authority	Co-presence, immediacy, perceived attentiveness	AI interpreted as instructor rather than assistant	Pedagogical authority
Engagement vs. Over-reliance	Sustained attention and motivation	Learner dependency and reduced productive struggle	Epistemic agency
Visibility vs. Accountability	Legible and explicit AI presence	Diffused responsibility across sociotechnical systems	Moral accountability
Augmentation vs. Replacement	Scalable support for educators	Normalization of delegated teaching roles	Role of human educators

perceptual. In classrooms, presence is historically entangled with authority. For example, in XR educational systems that include human-like AI avatars positioned alongside instructional materials, such as XR4ED [37], learners may interpret the agent as occupying a teacher-like role regardless of designers’ intentions.

Embodiment does not only add social cues; it also introduces classroom staging. When an agent is positioned “at the front”, granted the conversational floor, or allowed to initiate turns, it inherits interactional privileges that learners commonly associate with instruction. In this way, authority can emerge from spatial placement and interaction choreography. Even if the agent’s role is described as “assistive”, its embodied participation makes it legible as the party who explains and sets the terms of understanding [1].

This raises a critical question. When AI is present, who is perceived as responsible for learning? Embodied presence may unintentionally displace human educators as epistemic anchors, not through explicit replacement but through symbolic positioning. The tension, then, is not between presence and absence, but between presence and legitimacy. In other words, when does “being there” become “being in charge”?

4.2 Engagement vs. Over-reliance

Embodied AI systems are often justified through their capacity to increase engagement and motivation, particularly in settings where attention is scarce or uneven [16]. Socially expressive agents can encourage persistence, scaffold interaction, and reduce perceived effort. These benefits, however, come with a cost. In platforms that offer always-available AI tutors embedded in immersive environments, such as VictoryXR [35], sustained engagement may come at the cost of reduced productive struggle or peer interaction.

Embodied AI may also discourage productive struggle, peer collaboration, or critical questioning [9]. When guidance is always available (and likely delivered by an entity that appears attentive and authoritative) learners may defer to explore, or comply rather than challenge. Over time, this can shift learning from an active process of sense-making to a passive process of following [4].

This tension complicates dominant narratives of “support”. At what point does embodied assistance become pedagogical dependency? And how might engagement metrics obscure longer-term effects on autonomy, confidence, and epistemic agency?

4.3 Visibility vs. Accountability

An argument in favor is that embodiment increases transparency. A visible agent makes AI’s presence explicit rather than hidden in the background. Yet visibility does not guarantee accountability.

While embodied agents in XR platforms make AI highly visible to learners, responsibility remains distributed across designers, developers, institutions, and datasets [7, 32]. When an embodied agent provides misleading guidance, reinforces bias, or fails to support a learner, accountability rarely attaches to the visible entity itself. Instead, it dissolves into sociotechnical infrastructure.

This creates a troubling asymmetry. Embodiment concentrates authority while diffusing responsibility. Learners encounter a body that appears present and intentional, yet there is no clear locus of accountability for its actions. This tension challenges assumptions that making AI visible is sufficient for responsible deployment [33].

4.4 Augmentation vs. Replacement

AI in education is frequently framed as a means to support educators instead not replacing them [14, 16, 36]. However, systems that frame embodied AI as “virtual teachers” or full-time tutors implicitly rehearse futures in which instructional roles are partially delegated, even when marketed as supportive or supplemental.

By giving AI a body that explains, evaluates, and guides, designers implicitly model roles traditionally held by humans. Even when framed as assistance, these roles can gradually expand. Over time, what begins as augmentation may normalize delegation [6].

This tension highlights the performative role of design. Embodied AI does not simply respond to present needs; it shapes expectations about who (or what) teaches. The question is not whether replacement is intended, but whether it is being quietly rehearsed.

5 Why These Tensions Persist

A natural question follows from the tensions. If the risks of embodied AI in education are so substantial, why do these patterns keep recurring? Why does embodiment repeatedly emerge as a desirable design direction, even in contexts where authority, responsibility, and care are already heavily contested?

One reason is that embodiment aligns closely with dominant evaluative practices in HCI and educational technology. Embodied systems are comparatively easy to demonstrate, prototype, and assess through short-term studies focused on engagement, presence, or perceived usefulness. Prior work has shown that such

metrics privilege immediacy and experiential salience over longer-term questions of legitimacy, accountability, and institutional impact [3, 29]. As a result, embodiment is rewarded not because it is pedagogically appropriate, but because it is empirically legible within prevailing research paradigms.

A second reason lies in institutional narratives around scalability and personalization. AI in education is frequently justified as a response to systemic pressures. Large class sizes, limited resources, and increasing demands for individualized support create pressures that make AI-based assistance appear attractive [13, 16]. Embodied AI offers a compelling vision in this context, through for example, personalized guidance delivered at scale [25]. However, as scholars often note, such narratives may obscure the normative implications of delegating pedagogical roles to automated systems, framing structural challenges as technical opportunities [36].

Third, embodiment resonates with long-standing commitments in HCI to make interaction more “natural,” intuitive, and human-like. These commitments are deeply embedded in the field’s history and are often treated as self-evident goods [8]. Yet in educational contexts, naturalness can be misleading. Teaching is not an everyday social interaction, but an institutionalized practice governed by norms of authority, care, and accountability. Designing AI to participate in this practice as if it were merely another social actor risks collapsing critical distinctions between assistance and instruction, support and authority [32].

Finally, responsibility for the effects of embodied AI is structurally diffuse. While authority is enacted through a visible agent, accountability is distributed across designers, developers, institutions, and policies [5]. Prior work in responsible AI highlights how such diffusion makes it difficult to contest or resist problematic design trajectories, even when harms are anticipated [7]. In educational settings, this diffusion allows embodied AI to appear benign or inevitable, while its authority effects accumulate incrementally.

Taken together, these dynamics offer a lens for understanding why the tensions recur across embodied AI efforts in education. They are not failures of individual systems or designers, but symptoms of broader evaluative, institutional, and disciplinary logics. Recognizing this persistence is a necessary step toward treating embodiment not as a default design trajectory, but as a choice that warrants debate, constraint, and, in some cases, refusal.

6 Questioning the Right to Have a Body

The preceding analysis reframes embodied AI in education as more than a design challenge. It positions embodiment as a normative act; one that authorizes participation, signals legitimacy, and reshapes responsibility within educational settings. From this perspective, the central issue is not how embodied AI should be designed, but whether (and under what conditions) it should be granted bodily presence in the first place.

This reframing shifts the locus of responsibility. It asks HCI to move beyond optimization toward questions of legitimacy, boundary-setting, and refusal. Not every technically feasible design direction warrants pursuit, particularly in environments as value-laden as education. When AI is given a body, design decisions no longer concern interaction quality alone; they concern who is authorized to teach, guide, and evaluate.

Seen this way, embodied AI exposes the limits of a dominant assumption in HCI, that is, *risks introduced by new technologies can be adequately mitigated through transparency, better interfaces, or careful calibration*. Once authority is enacted through bodily presence, it cannot be fully constrained through disclaimers, explanations, or user awareness. Instead, it reshapes interpretation and expectation. This shift has direct implications for how embodied AI systems are framed, studied, and justified. Questions such as “Does this agent improve engagement?” or “Do learners trust it?” become insufficient on their own [28]. Instead, as an HCI community, we need to move more difficult questions to the foreground. “Who is positioned as responsible when guidance fails?” “What forms of authority are being normalized through design?” “Which educational roles are quietly being rehearsed as delegable?”

By treating embodiment primarily as a benefit, HCI risks normalizing a redistribution of pedagogical authority without adequate debate. With this provocation, we call for a shift in how embodied AI is judged and governed; from interface enhancement to institutional intervention, from engagement to legitimacy, and from what AI can do to what AI should be allowed to become. When AI has a body, it teaches more than content. It teaches norms about authority, expertise, and responsibility. The question is whether HCI is prepared to take responsibility for those lessons.

7 Conclusion

Embodied AI in education is often discussed as a matter of interaction quality, engagement, or technological progress. We argue that such framings are incomplete. When AI systems are given bodies in educational contexts, they do not merely support learning; they participate in the social and moral organization of teaching. By articulating the embodiment trap and a set of tensions that resist optimization, this provocation invites the HCI community to treat embodiment not as a default design trajectory, but as a normative intervention that warrants scrutiny, debate, and restraint.

Although our argument centers on education, the embodiment trap extends to XR training and operational environments in Industry 5.0 and human–robot collaboration. Adopting this perspective in pilot deployments would allow evaluations to move beyond ergonomic efficiency and task performance, and examine normative consequences such as worker autonomy and accountability in AI-assisted decision-making, and the risk of over-reliance on embodied XR. For example, personalized remote maintenance or assembly guidance could explicitly signal uncertainty and preserve human supervisory control, and be evaluated based on long-term skill retention rather than short-term task completion. Similarly, XR training modules could emphasize collaborative co-learning rather than authoritative instruction. They could include explainable-AI cues and assess long-term human expertise development alongside task performance. Adopting the embodiment-trap perspective as an evaluation lens ensures that XR systems remain aligned with principles of human-centric AI while preventing unintended shifts in epistemic authority in training and decision-support contexts.

We hope this work contributes to ongoing conversations about responsible AI, not by offering solutions, but by clarifying what is at stake when AI is allowed to take up space in the classroom.

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References

- [1] Tony Belpaeme, James Kennedy, Aditi Ramachandran, Brian Scassellati, and Fumihide Tanaka. 2018. Social robots for education: A review. *Science robotics* 3, 21 (2018), eaat5954. <https://doi.org/10.1126/scirobotics.aat5954>
- [2] Dan Bennett, Oussama Metatla, Anne Roudaut, and Elisa D Mekler. 2023. How does HCI understand human agency and autonomy?. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–18. <https://doi.org/10.1145/3544548.3580651>
- [3] Frank Biocca, Chad Harms, and Judee K Burgoon. 2003. Toward a more robust theory and measure of social presence: Review and suggested criteria. *Presence* 12, 5 (2003), 456–480.
- [4] Michelene TH Chi and Ruth Wylie. 2014. The ICAP framework: Linking cognitive engagement to active learning outcomes. *Educational psychologist* 49, 4 (2014), 219–243. <https://doi.org/10.1080/00461520.2014.965823>
- [5] Marios Constantinides, Edyta Bogucka, Daniele Quercia, Susanna Kallio, and Mohammad Tahaei. 2024. RAI guidelines: Method for generating responsible AI guidelines grounded in regulations and usable by (non-) technical roles. *Proceedings of the ACM on Human-Computer Interaction* 8, CSCW2 (2024), 1–28. <https://doi.org/10.1145/3686927>
- [6] Marios Constantinides and Daniele Quercia. 2025. AI, Jobs, and the Automation Trap: Where Is HCI?. In *Proceedings of the 4th Annual Symposium on Human-Computer Interaction for Work*. 1–8. <https://doi.org/10.1145/3729176.3729191>
- [7] Virginia Dignum. 2019. *Responsible artificial intelligence: how to develop and use AI in a responsible way*. Vol. 2156. Springer.
- [8] Paul Dourish. 2001. *Where the Action Is: The Foundations of Embodied Interaction*. MIT Press.
- [9] Tobias Drey, Patrick Albus, Simon der Kinderen, Maximilian Milo, Thilo Segsneider, Linda Chanzab, Michael Rietzler, Tina Seufert, and Enrico Rukzio. 2022. Towards collaborative learning in virtual reality: A comparison of co-located symmetric and asymmetric pair-learning. In *Proceedings of the ACM Conference on Human Factors in Computing Systems (CHI)*. 1–19. <https://doi.org/10.1145/3491102.3517641>
- [10] Engage XR. 2024. Engage XR. <https://engagevr.io/>. Accessed January 21, 2026.
- [11] Fectar. 2019. Fectar | Meet in AR Metaverse. <https://play.google.com/store/apps/details?id=io.spotn.viewer>. Accessed January 21, 2026.
- [12] Pascale Fung, Yoram Bachrach, Asli Celikyilmaz, Kamalika Chaudhuri, Delong Chen, Willy Chung, Emmanuel Dupoux, Hongyu Gong, Hervé Jégou, Alessandro Lazaric, Arjun Majumdar, Andrea Madotto, Franziska Meier, Florian Metz, Louis-Philippe Morency, Théo Moutakanni, Juan Pino, Basile Terver, Joseph Tighe, Paden Tomasello, and Jitendra Malik. 2025. Embodied AI Agents: Modeling the World. <https://doi.org/10.48550/arXiv.2506.22355>
- [13] Ariel Han, Xiaofei Zhou, Zhenyao Cai, Shenshen Han, Richard Ko, Seth Corrigan, and Kylie A Peppler. 2024. Teachers, parents, and students' perspectives on integrating generative AI into elementary literacy education. In *Proceedings of the ACM Conference on Human Factors in Computing Systems (CHI)*. 1–17. <https://doi.org/10.1145/3613904.3642438>
- [14] Emma Harvey, Allison Koenecke, and Rene F Kizilcec. 2025. "Don't Forget the Teachers": Towards an Educator-Centered Understanding of Harms from Large Language Models in Education. In *Proceedings of the ACM Conference on Human Factors in Computing Systems (CHI)*. 1–19. <https://doi.org/10.1145/3706598.3713210>
- [15] Cindy E. Hmelo-Silver. 2004. Problem-based learning: What and how do students learn? *Educational Psychology Review* 16, 3 (Sept. 2004), 235–266. <https://doi.org/10.1023/B:EDPR.0000034022.16470.f3>
- [16] Wayne Holmes. 2020. Artificial intelligence in education. In *Encyclopedia of education and information technologies*. Springer, 88–103.
- [17] Jean Lave and Etienne Wenger. 1991. *Situated learning: Legitimate peripheral participation*. Cambridge University Press.
- [18] Fotis Liarokapis, Václav Milata, Jose Luis Ponton, Nuria Pelechano, and Haris Zacharatos. 2024. XR4ED: An Extended Reality Platform for Education. *IEEE Computer Graphics and Applications* 44, 4 (2024), 79–88. <https://doi.org/10.1109/MCG.2024.3406139>
- [19] Fotis Liarokapis, Václav Milata, and Filip Skola. 2024. Extended reality educational system with virtual teacher interaction for enhanced learning. *Multimodal Technologies and Interaction* 8, 9 (2024), 83. <https://doi.org/10.3390/mti8090083>
- [20] Fotis Liarokapis, Václav Milata, and Filip Skola. 2024. Extended Reality Educational System with Virtual Teacher Interaction for Enhanced Learning. *Multimodal Technologies and Interaction* 8, 9 (2024). <https://doi.org/10.3390/mti8090083>
- [21] Florence Martin, Tong Wu, Liyong Wan, and Kui Xie. 2022. A Meta-Analysis on the Community of Inquiry Presences and Learning Outcomes in Online and Blended Learning Environments. *Online Learning* (2022). <https://doi.org/10.24059/olj.v26i1.2604>
- [22] Rozalind G. Muir-Herzig. 2004. Technology and its impact in the classroom. *Computers & Education* 42, 2 (Feb. 2004), 111–131. [https://doi.org/10.1016/S0360-1315\(03\)00067-8](https://doi.org/10.1016/S0360-1315(03)00067-8)
- [23] Mengxu Pan, Alexandra Kitson, Hongyu Wan, and Mirjana Prpa. 2025. Ellma-t: an embodied llm-agent for supporting english language learning in social vr. In *Proceedings of the ACM Designing Interactive Systems (DIS) Conference*. 576–594. <https://doi.org/10.1145/3715336.3735786>
- [24] Rock Yuren Pang, Hope Schroeder, Kynnedi Simone Smith, Solon Barocas, Ziang Xiao, Emily Tseng, and Danielle Bragg. 2025. Understanding the LLM-ification of CHI: Unpacking the Impact of LLMs at CHI through a Systematic Literature Review. In *Proceedings of the ACM Conference on Human Factors in Computing Systems (CHI)*. 1–20. <https://doi.org/10.1145/3706598.3713726>
- [25] Hyanghee Park and Daehwan Ahn. 2024. The promise and peril of ChatGPT in higher education: opportunities, challenges, and design implications. In *Proceedings of the ACM Conference on Human Factors in Computing Systems (CHI)*. 1–21. <https://doi.org/10.1145/3613904.3642785>
- [26] Jocelyn Parong, Kimberly A. Pollard, Benjamin T. Files, Ashley H. Oiknine, A. Sinatra, Jason D. Moss, Antony D. Passaro, and P. Khooshabeh. 2020. The mediating role of presence differs across types of spatial learning in immersive technologies. *Computers in Human Behavior* 107 (2020), 106290. <https://doi.org/10.1016/j.chb.2020.106290>
- [27] Rolf Pfeifer and Fumiya Iida. 2004. Embodied Artificial Intelligence: Trends and Challenges. In *Embodied Artificial Intelligence: International Seminar, Dagstuhl Castle, Germany, July 7–11, 2003. Revised Papers*. Springer, Berlin, Heidelberg, 1–26. https://doi.org/10.1007/978-3-540-27833-7_1
- [28] Snehal Prabhudesai, Ananya Prashant Kasi, Anmol Mansingh, Anindya Das Antar, Hua Shen, and Nikola Banovic. 2025. "Here the GPT made a choice, and every choice can be biased": How Students Critically Engage with LLMs through End-User Auditing Activity. In *Proceedings of the ACM Conference on Human Factors in Computing Systems (CHI)*. 1–23. <https://doi.org/10.1145/3706598.3713714>
- [29] Byron Reeves and Clifford Nass. 1996. *The Media Equation: How People Treat Computers, Television, and New Media Like Real People and Places*. CSLI Publications and Cambridge University Press.
- [30] Judy Robertson, Andrew Macvean, and Katy Howland. 2012. Embedding technology in the classroom: the train the teacher model. In *Proceedings of the ACM International Conference on Interaction Design and Children (IDC)*. ACM, 20–29. <https://doi.org/10.1145/2307096.2307100>
- [31] Neil Selwyn. 2019. *Should robots replace teachers?: AI and the future of education*. John Wiley & Sons.
- [32] Lucille Alice Suchman. 2007. *Human-machine reconfigurations: Plans and situated actions*. Cambridge university press.
- [33] Mohammad Tahaei, Marios Constantinides, Daniele Quercia, Sean Kennedy, Michael Muller, Simone Stumpf, Q Vera Liao, Ricardo Baeza-Yates, Lora Aroyo, Jess Holbrook, et al. 2023. Human-centered responsible artificial intelligence: Current & future trends. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–4. <https://doi.org/10.1145/3544549.3583178>
- [34] Janice G. Thorpe and Sherwyn P. Morreale. 2023. Immediacy and presence: reflections on an interdisciplinary and rhetorical approach to teaching and learning in the online environment. *Review of Communication* 23, 4 (2023), 348–366. <https://doi.org/10.1080/15358593.2023.2186801>
- [35] VictoryXR. 2023. VictoryXR. <https://www.victoryxr.com/>. Accessed January 21, 2026.
- [36] Ben Williamson and Rebecca Eynon. 2020. Historical threads, missing links, and future directions in AI in education. , 223–235 pages. <https://doi.org/10.1080/17439884.2020.1798995>
- [37] XR4ED Project. 2023. XR4ED: Extended Reality Platform for Education. <https://xr4ed.eu/>. Accessed January 21, 2026.
- [38] Erin J Zaffini. 2018. Communities of practice and legitimate peripheral participation: A literature review. *Update: Applications of Research in Music Education* 36, 3 (2018), 38–43. <https://doi.org/10.1177/8755123317743977>
- [39] Tianyi Zhang, Colin Au Yeung, Emily Aurelia, Yuki Onishi, Neil Chulpongsatorn, Jiannan Li, and Anthony Tang. 2025. Prompting an Embodied AI Agent: How Embodiment and Multimodal Signaling Affects Prompting Behaviour. In *Proceedings of the ACM Conference on Human Factors in Computing Systems (CHI)*. ACM, 1–25. <https://doi.org/10.1145/3706598.3713110>
- [40] Zhuoran Zhao, Zhizhuo Yin, Jia Sun, and Pan Hui. 2024. Embodied AI-guided interactive digital teachers for education. In *SIGGRAPH Asia 2024 Educator's Forum*. 1–8. <https://doi.org/10.1145/3680533.3697070>